2024 MIT Science Bowl

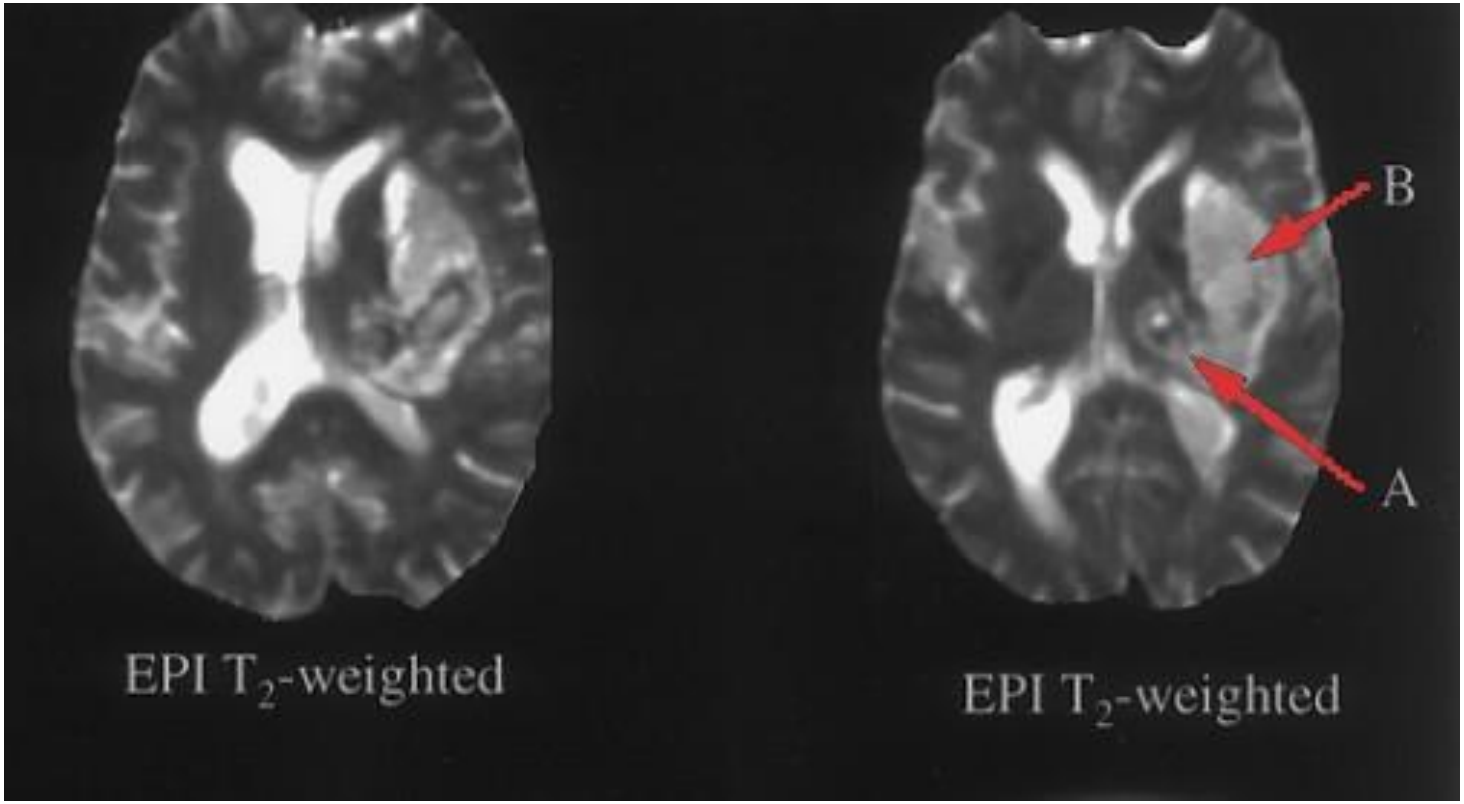
Round 13



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Bonus 2 (Bio)

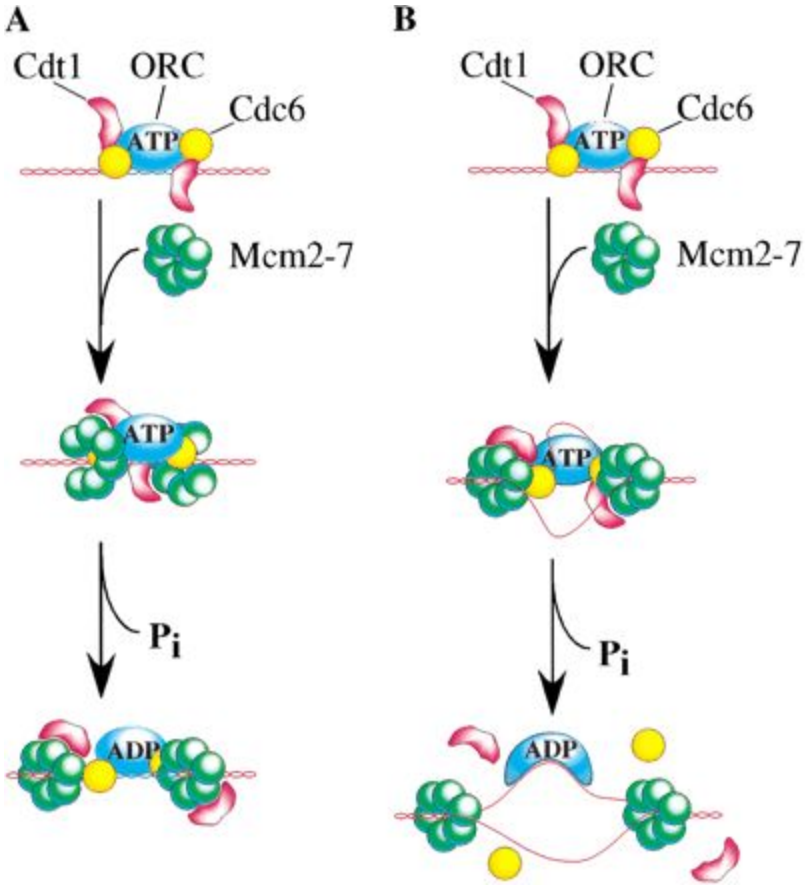

B
A
EPI T_2-weighted
EPI T_2-weighted



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Bonus 4 (Energy)

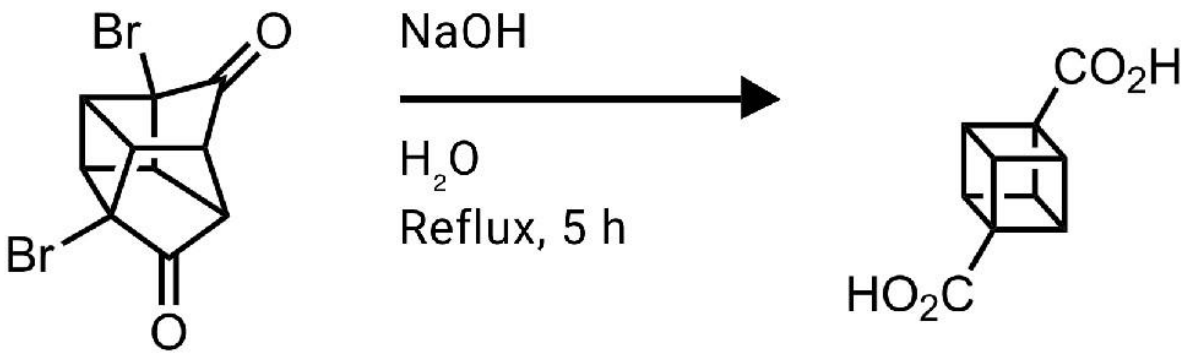

A
Cdt1
ORC
Cdc6
ATP
Mcm2-7
ATP
Pi
ADP
B
Cdt1
ORC
Cdc6
ATP
Mcm2-7
ATP
Pi
ADP



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Bonus 5 (Chem)

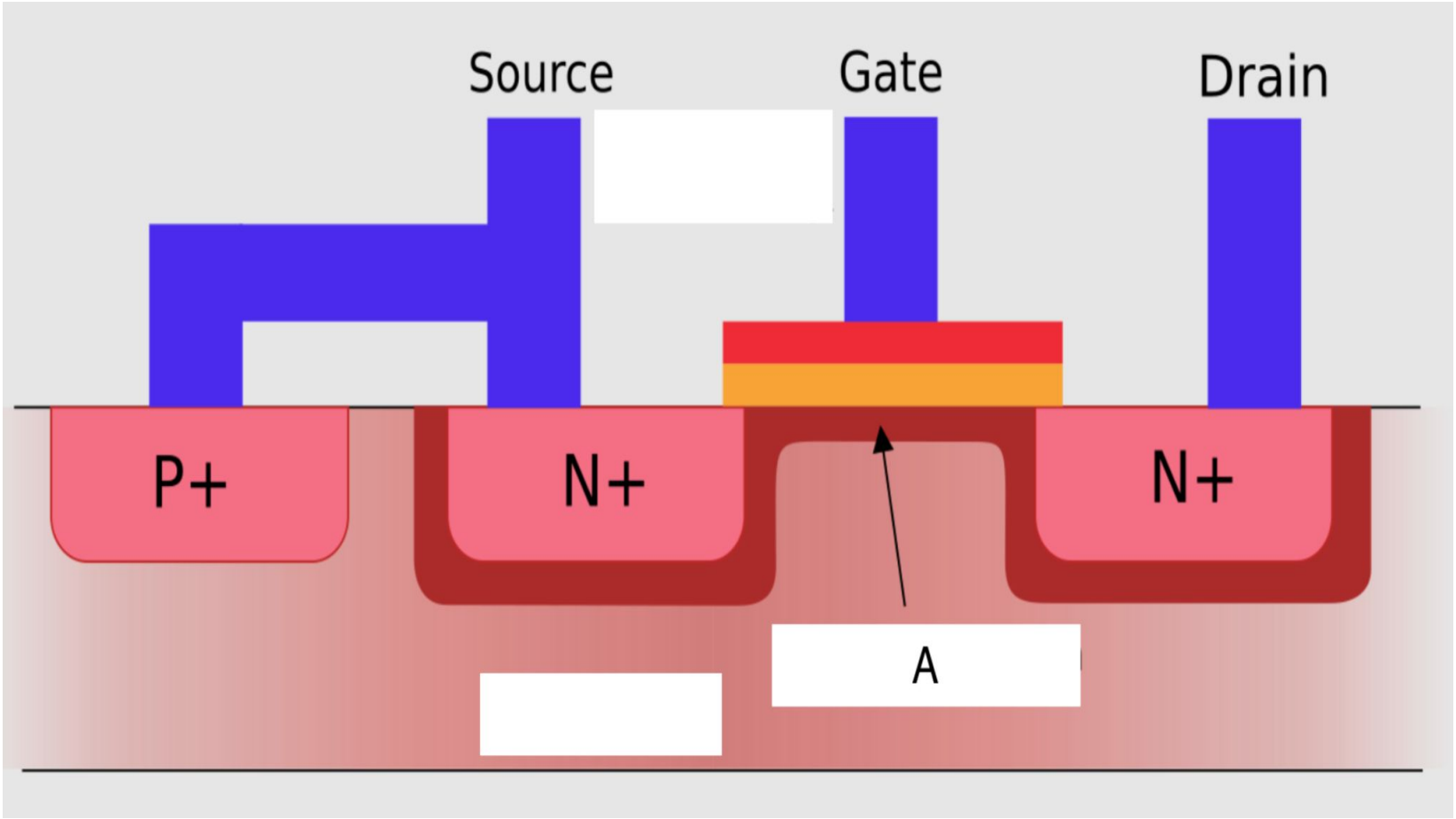

Br
O
Br
O
NaOH
H2O
Reflux, 5 h
CO2H
HO2C



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Bonus 6 (Physics)

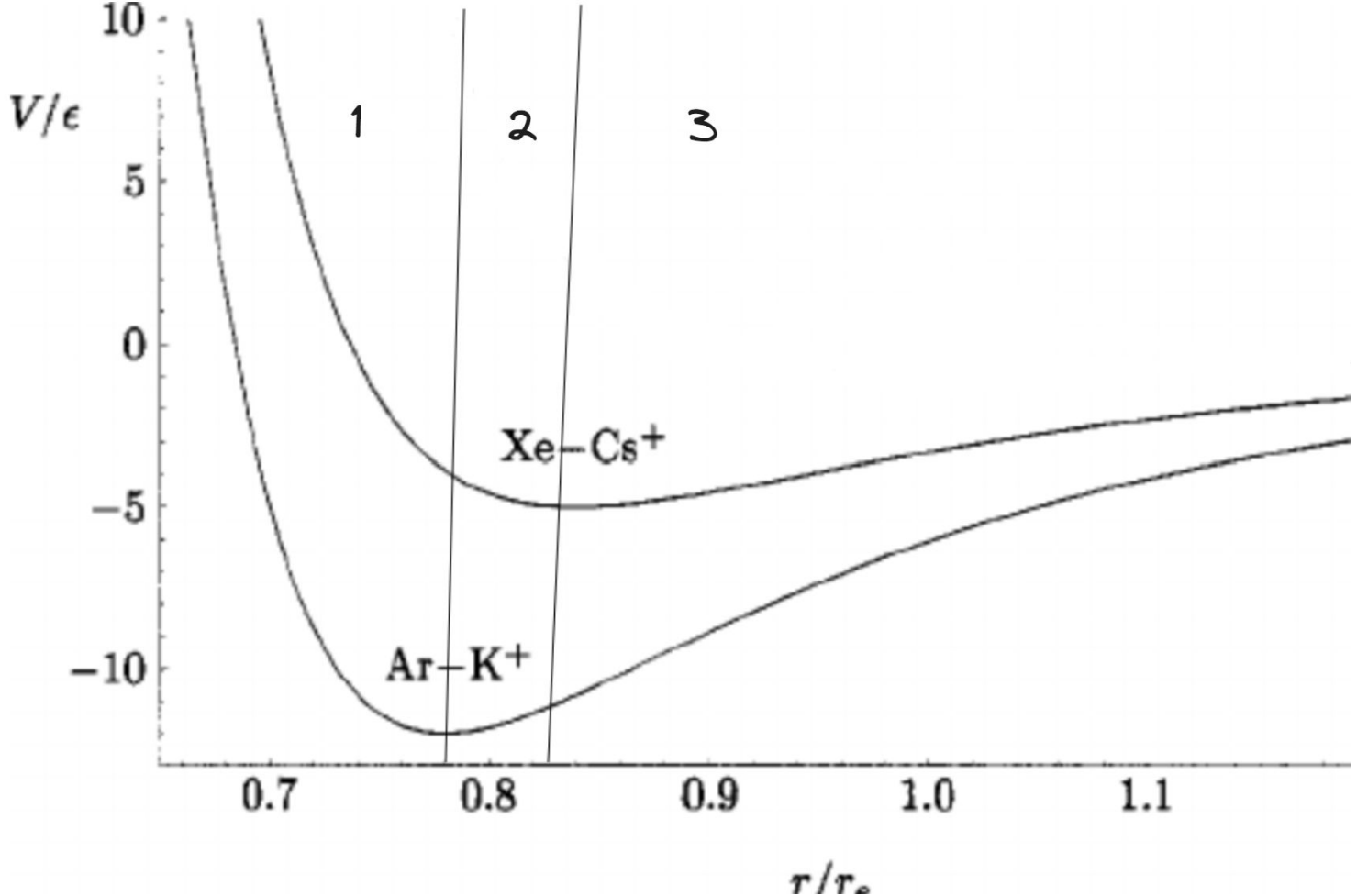

Source
Gate
Drain
P+
N+
N+
A



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Bonus 9 (Chem)

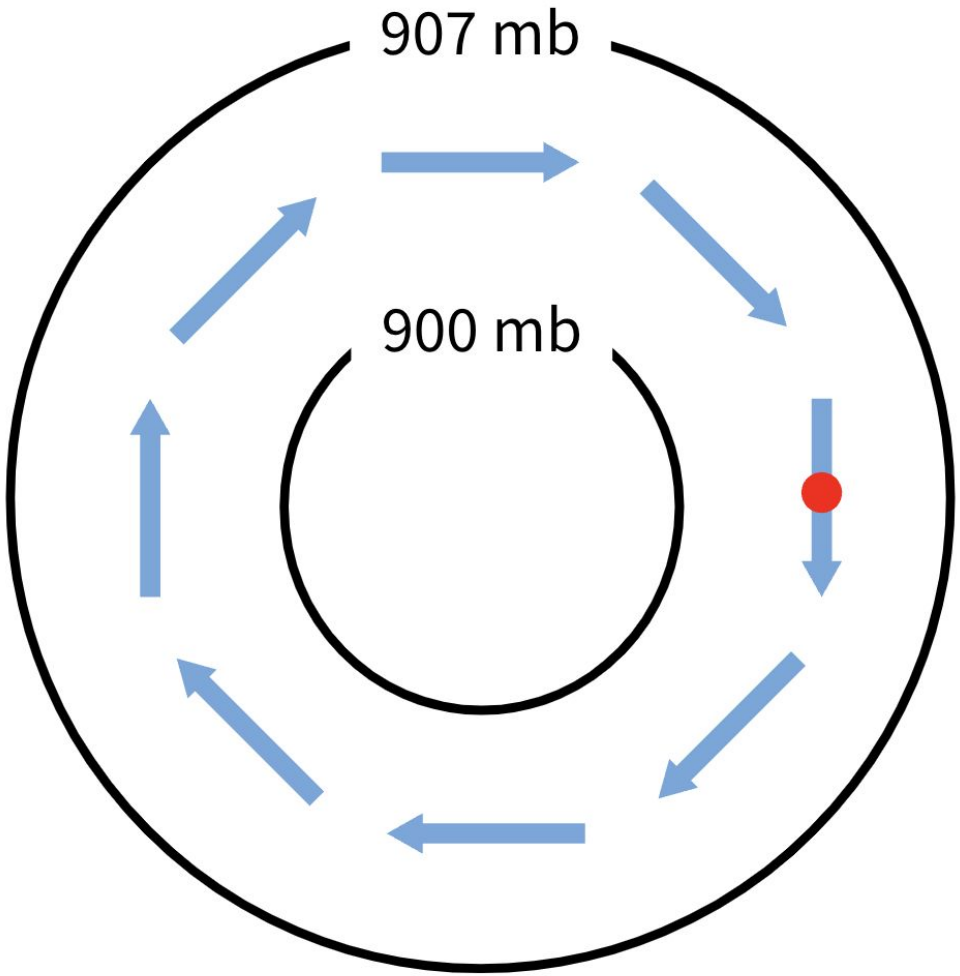

10
V/ϵ
5
0
−5
−10
1
2
3
Xe–Cs+
Ar–K+
0.7
0.8
0.9
1.0
1.1
r/r



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Bonus 10 (ESS)

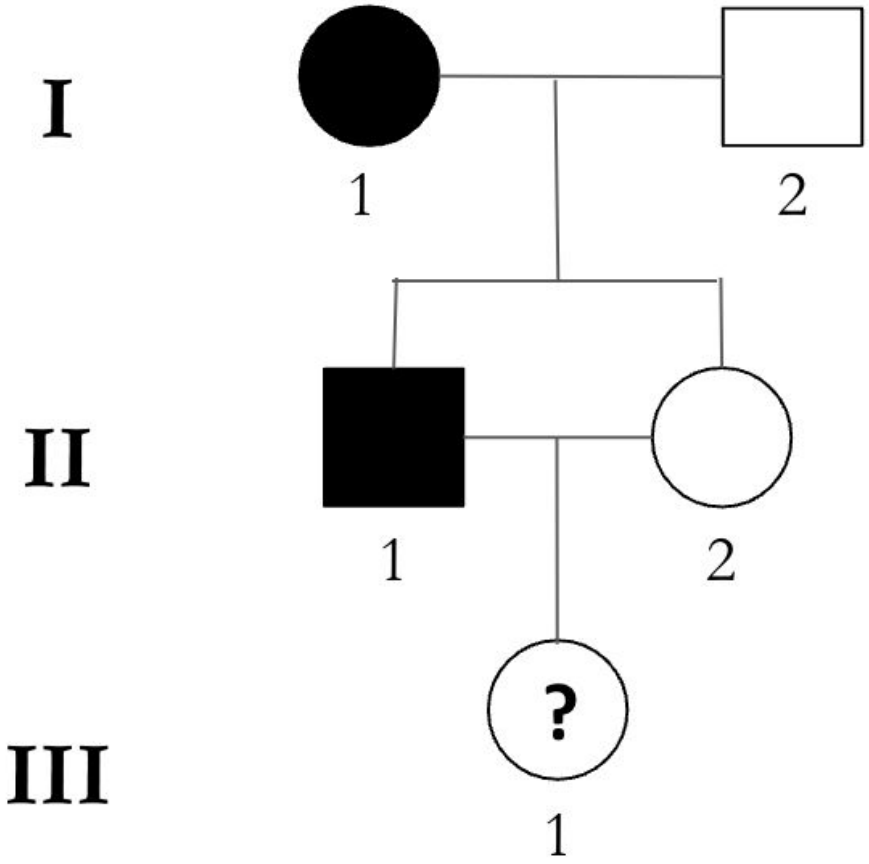

907 mb
900 mb



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Bonus 15 (Bio)

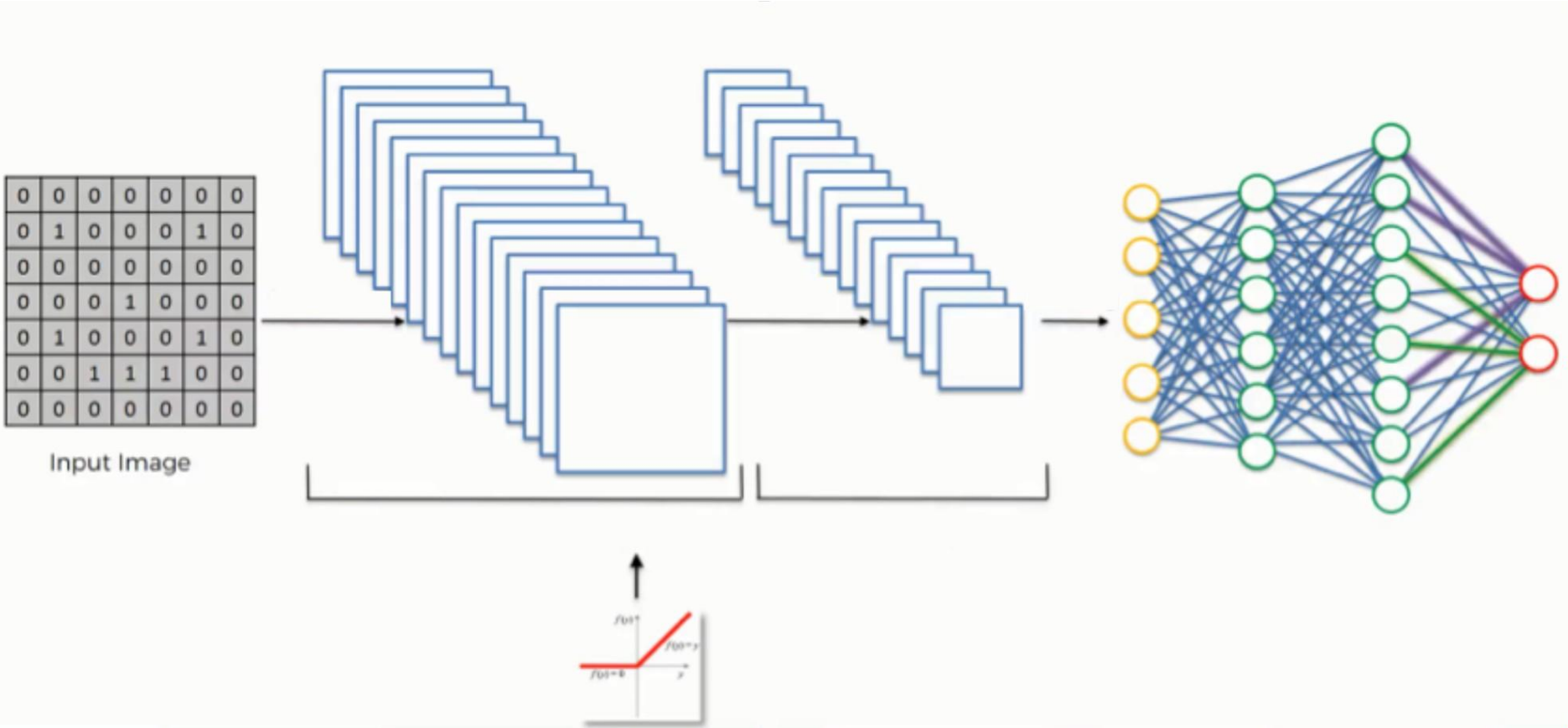

I
1
2
II
1
2
?
III
1



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a competition like NSB . . .

Bonus 17 (Energy)


0 0 0 0 0 0 0
0 1 0 0 0 1 0
0 0 0 0 0 0 0
0 0 0 1 0 0 0
0 1 0 0 0 1 0
0 0 1 1 1 0 0
0 0 0 0 0 0 0
Input Image



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Bonus 21 (Math)





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